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## TEST TIME ON CASELET PROBLEMS

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# TIME AND WORK

# (WORK WITH DIFFERENT EFFICIENCIES)

# TIME AND WORK (WORK WITH DIFFERENT EFFICIENCIES)

**Definition:**

Work is defined as something which has an effect or outcome; often the one desired or expected. The basic concept of Time and Work is similar to that across all Arithmetic topics, i.e. the concept of Proportionality.

Efficiency is inversely proportional to the Time taken when the amount of work done is constant.

$$\text{Efficiency} \propto \frac{1}{\text{Time Taken}}$$

The complete work can also be considered as 1 unit. Then if A takes 4 days to finish a work, it means he can finish 1/4th of the work in 1 day.

**Time and Work Shortcut Tricks**:

| Number of days | Percentage value |
|---|---|
| 2 | 50% |
| 3 | 33% |
| 4 | 25% |
| 5 | 20% |
| 6 | approx.16.67% |
| 7 | approx.14% |
| 8 | approx.12% |
| 9 | approx.11% |

Approach 1: Using Fractions
Ram can finish the work in 10 days i.e. in one day he will do 1/10th of the work.
Rahim can finish the work in 40 days i.e. in one day he will do 1/40th of the work.
So, in one day, both working together can finish= (1/10) + (1/40) = 5/40 = 1/8th of the work. So, to complete the work they will take 8 days.

Approach 2: Using Percentage
Rahim can finish 100 % of work in 10 days i.e. in one day he finishes 10% of the work.
Ram can finish 100% of the work in 40 days i.e. in one day he finishes 2.5 % of the work.
So, working together, in a single day they can finish 12.5% of the work. So, to complete 100% of the work, they will take 100/12.5 = 8 days.

Work from Days:
If A can do a piece of work in n days, then A's 1 day's work = 1/ n.

Days from Work:

If A's 1 day's work = 1/ n, then A can finish the work in n days.

Ratio:

If A is thrice as good a workman as B, then:
Ratio of work done by A and B = 3 : 1.
Ratio of times taken by A and B to finish a work = 1 : 3.

## Question: 01

A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

A. 12 days
B. 15 days
C. 16 days
D. 18 days

**Answer: B**

# Explanation: 01

A’s 2 day’s work = (1/ 20)*2 = 1/ 10

(A+B+C)’s 1 day’s work = (1/ 20)+(1/ 30)+(1/ 60)

= 6/ 60 = 1/10

Work done in 3 days = (1/ 10)+(1/ 10)

= 1/ 5

Now, 1/ 5 work is done in 3 days.

Therefore, Whole work will be done in (3*5) =15 days.

Hence, option B is correct.

## Question: 02

A can lay railway track between two given stations in 16 days and B can do the same job in 12 days. With help of C, they did the job in 4 days only. Then, C alone can do the job in:

A. 46/ 5
B. 47/5
C. 48/5
D. 10

**Answer: C**

## Explanation: 02

(A+B+C)’s 1 day work = 1/ 4

A’s 1 day work = 1/ 16

B’s 1 day work = 1/ 12

Therefore, C’s 1 day work = [(1/ 4) - (1/ 16)+(1/ 12)]

= (1/ 4) - (7/ 48) = 5/ 48

So C alone can do the work in 48/ 5.

Hence , option C is correct.

## Question: 03

Aadi can do a piece of work in 12 days alone, Bunny can do the same work in 16 days alone. After Aadi has been working for 5 days and Bunny for 7 days, Charan finishes it in 14 days. In how many days will Charan alone be able to do the work?

A. 96 days
B. 48 days
C. 24 days
D. 12 days

**Answer: A**

# Explanation: 03

According to the given data,

$5/12 + 7/16 + 14/d = 1$

or, $14/d = 1 - (5/12 + 7/16)$

or, $14/d = 1 - (20+21)/48$

or, $14/d = (48 - 41)/48$

or, $2/d = 1/48$

or, $d = 2\times48 = 96$ days

Therefore, Charan alone can complete the work in 96 days.

Hence, option A is work.

## Question: 04

Kaushalya can do a work in 20 days, while kaikeyi can do the same work in 25 days. They started the work jointly Few days later Sumitra also joined them and thus all of them completed the whole work in 10 days. All of them were paid total Rs.700. What is the Share of Sumitra?

A. Rs.130
B. Rs.185
C. Rs.70
D. can't be determined

**Answer: C**

# Explanation: 04

Efficiency of kaushalya = 5%

Efficiency of kaikeyi = 4%

Thus, in 10 days working together they will complete only 90% of the work.

So, [(5+4)*10] =90

Hence, the remaining work will surely done by sumitra, which is 10%.

Thus, sumitra will get 10% of Rs. 700, which is Rs.70

Hence, option C is correct.

## Question: 05

When A, B and C are deployed for a task , A and B together do 70% of the work and B and C together do 50% of the work. who is most efficient?

A. A
B. B
C. C
D. Can't be determined

**Answer: A**

# Explanation: 05

A + B = 0.7 W

B + C = 0.5 W

A+B+C = 1

C = 1 - 0.7 = 0.3

B = 0.5 - 0.3 = 0.2

A = 0.7 - 0.2 = 0.5

So A is most efficient.

Hence, option A is correct.

## Question: 06

A does half as much work as B and C does half as much work as A and B together. If C alone can finish the work in 40 days, then together ,all will finish the work in:

A. 17 + 4/7 days
B. 13 + 1/3 days
C. 15 + 3/2 days
D. 16 days

**Answer: B**

# Explanation: 06

C alone can finish the work in 40 days.

As given C does half as much work as A and B together

=> (A + B) can do it in 20 days

(A + B)’s 1 days wok = 1/20.

A's 1 days work : B's 1 days Work = 1/2 : 1 = 1:2(given)

A's 1 day’s work = (1/20) x (1/3) = (1/60) [Divide 1/20 in the ratio 1:2]

B's 1 days work = (1/20) x (2/3) = 1/30

(A+B+C)'s 1 day's work = (1/60) + (1/30) + (1/40) = 9/120 = 3/40

All the three together will finish it in 40/3 = 13 and 1/3 days.

Hence, option B is correct.

# Question: 07

A, B and C can do a job in 6 days. 12 days and 15 days respectively. After 1/ 8 of the work is completed, C leaves the job. Rest of the work is done by A and B together, Time taken to finish the remaining work is

A. 7/ 2 Days
B. 9/ 2 Days
C. 17/ 2 Days
D. 13/ 2 Days

**Answer: A**

# Explanation: 07

1/ 8 of work  = 60/ 8 = 15/ 2 units.

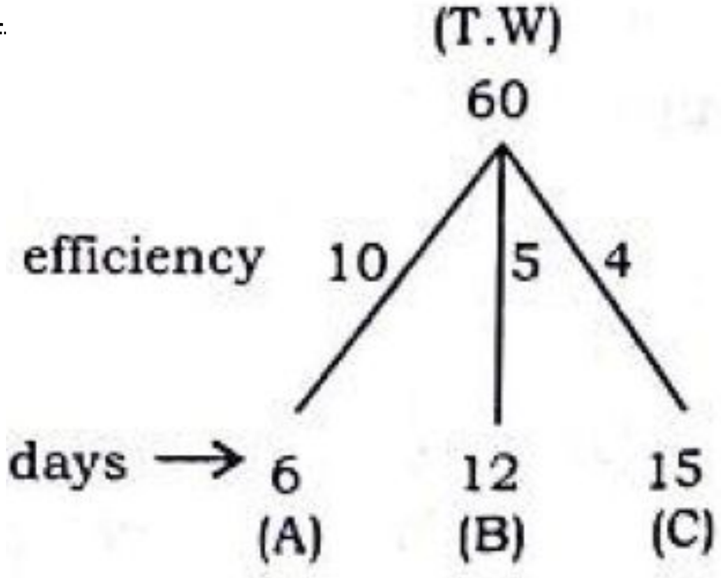
Rest work = 60-(15/ 2) = 105/ 2 units

Rest of work completed by A+B = Rest work/ (A+B)'s ef

= 105/ (2*15)

= 7/ 2.

Hence, option A is correct.



# Question: 08

Among four persons Prince, Queen, Raj and Sashi. Prince takes thrice as much time as Queen to complete a piece of work. Queen takes thrice as much time as Raj and Raj takes thrice as much time as Sashi to complete the same work. One group of three of the four men can complete the work in 13 days while another group of three can do so in 31 days. Which is the group that takes 13 days?

A. Prince , Queen , Raj
B. Prince ,Queen , Sashi
C. Queen , Raj , Sashi
D. Prince , Raj , Sashi

**Answer: B**

# Explanation: 08

From the given information Q is thrice as efficient as P.
R is thrice as efficient as Q.
S is thrice efficient as R.
If work done by P in a day is 'n' units, the work done in a day by Q, R and S would be 3n, 9n and 27n units respectively.
It can be seem that, P,Q and R working together can do 13n units in a day while P,Q and S working together can do 31n units in a day.
Hence, the ratio of times taken to complete the work by the former and later groups is 31:13
P,Q and S take 13 days.
Hence, option B is correct.

## Question: 09

Rama can do a piece of work in 10 days and Rami alone can do it in 5 days. Rama and Rami undertook to do it for Rs.400. With the help of their neighbour Snooty, they finished it in 2 days. The share of Snooty out of the remuneration is

A. Rs.120
B. Rs.200
C. Rs.160
D. Rs.180

**Answer: C**

# Explanation: 09

(Rama + Rami + Snooty)'s one day's work = 1/10 + 1/5 + 1/X.
(Rama + Rami + Snooty)'s two day's work = 2 x (1110 + 1/5 + 1/X).
{because they need 2 days to do the job)
=> 2 [ 1/10 + 115 + 1/X ] = 1 = On solving we get, X = 5 days.
This means that Snooty can do the whole work alone in 5 days.
=> In one day, he does 1/5 of the work.
=>In two days, he does (2 x 1/5) of the work.
=> Snooty's share = 1/5 x 2 x 400 = Rs.160.
Hence, option C is correct.

## Question: 10

Billy is four times as good a workman as Silly and can build a wall in 45 days less than the number of days required by Silly. What is the time they will take, working together, to build two such walls?

A. 12 days
B. 24 days
C. 9 days
D. 18 days

**Answer: B**

# Explanation: 10

Let Billy's time to complete one work = x days.
Then, Silly's time to complete one work = 4x days.
Since Billy take 45 days less than Silly to build a wall.
i.e., $4x - x = 45 \Rightarrow x = 15$ and $4x = 60$.
Working together they will take $1/(1/15+1/60)=12$ days to build one wall.
So, to build two such walls, 24 days will be required.
Hence, option B is correct.

# Question: 11

Two men undertake to do a piece of work for Rs.200. One alone could do it in 6 days, the other In 8 days. With the assistance of a boy they finish it in 3 clays. What Is the share of the boy?

A. Rs.25
B. Rs.50
C. Rs.60
D. Rs.30

**Answer: A**

# Explanation: 11

1st man's a days work = 3/6, 2nd man's 3 days work = 3/8.
The boy's 3 days' work = 1 — (3/6 + 3/8) = 1/8.
Clearly Rs.200 should be divided amongst them in the proportion of 3/6 : 3/8 : 1/8 or 4 : 3 : 1.
1st man's share = 4/8 of Rs.200 = Rs.100.
2nd man's share = 3/8 of Rs.200 = Rs.75.
The boy's share = 1/8 of Rs.200 = Rs.25.
Hence, option A is correct.

## Question: 12

Two coal loading machines each working 12 hours per day for 8 days handles 9 tons of coal with an efficiency of 90%. While 3 other coal loading machines at an efficiency of 80% set to handle 12 tons of coal In 6 days. Find how many hours per day each should work?

A. 12 hrs/day
B. 16 hrs/day
C. 20 hrs /day
D. 18 hrs/day

**Answer: B**

# Explanation: 12

$(N1 \times D1 \times H1 \times E1)/W1 = (N2 \times D2 \times H2 \times E2)/W2$ (By chain Rule).
$= (2 \times 8 \times 12 \times 90)/9 = (3 \times 6 \times H2 \times 80)/12 \Rightarrow H2 = 16$ hours/day
Hence, option B is correct.

## Question: 13

To do a certain piece of work, B would take three times as long as A and C together and C twice as long as A and B together. The three men working together can complete the work in 10 days. How long would B take by himself to complete the same piece of work?

A. 24 days
B. 30 days
C. 40 days
D. 36 days

**Answer: C**

# Explanation: 13

By the question, 3 times B's daily work = (A + C)'s daily work.
Add B's daily work to both sides.
=> 4 times B's daily work = (A + B + C)'s daily work = 1/10.
=>B's daily work = 1/40 .... (1)
Also, 2 times C's daily work = (A + B)'s daily work.
Add C's daily work to both sides.
=>3 times C's daily work = (A + B + C)'s daily work = 1/10.
=>C's daily work = 1/30 .... (2)
Now A's daily work = 1/10 — (1/40 + 1/30) = 1/24 ...(3)
=>A, B and C can do the work in 24, 40 and 30 days respectively.
Hence, option C is correct.

# Question: 14

Three factories of Conglomerate Corporation are capable of manufacturing hubcaps. Two of the factories can each produce 100,000 hubcaps in 15 days. The third factory can produce hubcaps 30% faster. Approximately how many days would it take to produce a million hubcaps with all three factories working simultaneously?

A. 38
B. 42
C. 46
D. 50

**Answer: C**

# Explanation: 14

since third factory is 30% faster , thus it will create 130,000 hubcaps in 15 days
in 15 days total number of hubcaps made by all three will be :
100,000+100,000+130,000 = 330,000
in 30 days:
330,000+330,000 = 660,000
in 45 days:
660,000+330,000 = 990,000
now only 10,000 more hubcaps are required , which will take 1 day
thus in all 46 days.
Hence, option C is correct.

## Question: 15

A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work?

A. 5
B. 6
C. 7
D. 8

**Answer: B**

## Explanation: 15

1. B works for 10 days and completes $\frac{10}{15} = \frac{2}{3}$ of the work.
2. Remaining work is $1 - \frac{2}{3} = \frac{1}{3}$.
3. A can complete $\frac{1}{3}$ of the work in $\frac{1}{3} \times 18 = 6$ days.

**Answer: 6 days** (option b)

## Question: 16

Two pipes A and B independently can fill a tank in 20 hours and 25 hours. Both are opened together for 5 hours after which the second pipe is turned off. What is the time taken by first pipe alone to fill the remaining portion of the tank?

A. 11 hours
B. 22 hours
C. 55 hours
D. 9 hours

**Answer: B**

# Explanation: 16

- ➔ Total unit water = LCM(20, 25) = 100 unit
- ➔ A’s efficiency = 100/20 = 5 unit/hour
- ➔ B’s efficiency =100/25 = 4 unit/hour
- ➔ After 5 hour the water filled by A and B together = 5 x 9 =45 unit
- ➔ Remaining unit = 100 – 45 = 55 unit
- ➔ Time taken by A alone = 55/5 = **11 hours**

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